

#### IN THIS CHAPTER

- » Getting a handle on networks
- » Considering why networking is useful (and is everywhere)
- » Telling the difference between servers and clients
- » Seeing how networks change computing life
- » Examining network topology
- » Identifying (and offering sympathy to) the network administrator

## Chapter 1

# Welcome to Networking

Computer networks get a bad rap in the movies. In the 1980s, the *Terminator* movies featured Skynet, a computer network that becomes self-aware (a computer network of the future), takes over the planet, builds deadly terminator robots, and sends them back through time to kill everyone unfortunate enough to have the name Sarah Connor. In the *Matrix* movies, a vast and powerful computer network enslaves humans and keeps them trapped in a simulation of the real world. And in the 2015 blockbuster *Spectre*, James Bond goes rogue (again) to prevent the Evil Genius Ernst Blofeld from taking over the world (again) by linking the computer systems of all the world's intelligence agencies together to form a single, all-powerful evil network that spies on everybody.

Fear not. These bad networks exist only in the dreams of science-fiction writers. Real-world networks are much more calm and predictable. Although sophisticated networks do seem to know a lot about you, they don't think for themselves and they don't evolve into self-awareness. And although they can gather a sometimes disturbing amount of information about you, they won't try to kill you, even if your name is Sarah Connor.

Now that you're over your fear of networks, you're ready to breeze through this chapter. It's a gentle, even superficial, introduction to computer networks, with a slant to the concepts that can help you use a computer that's attached to a network. This chapter goes easy on the details; the detailed stuff comes later.

## Defining a Network

A *network* is nothing more than two or more computers connected by a cable or by a wireless radio connection so that they can exchange information.

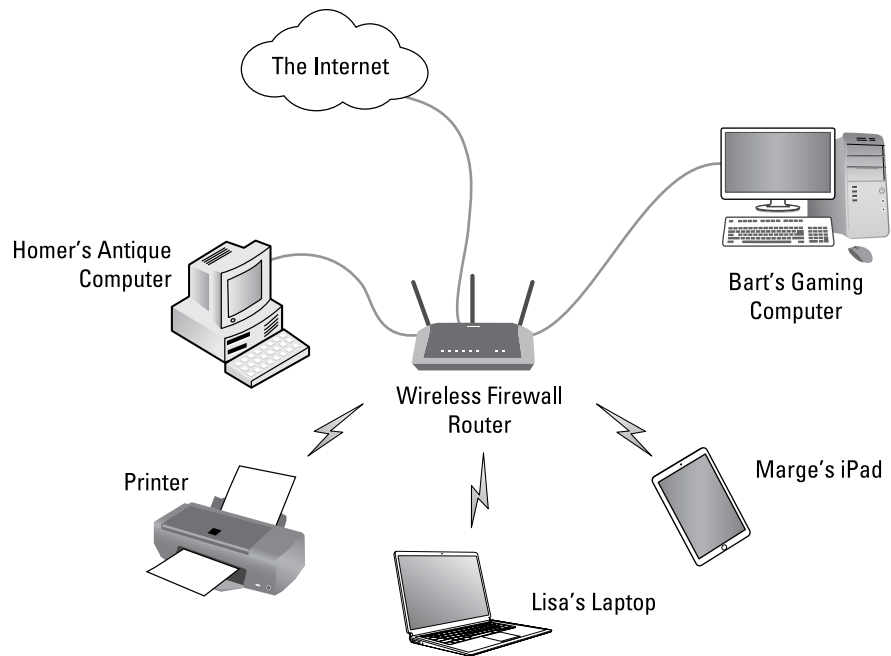
You can create a simple computer network by using a device called a *switch* to connect all the computers in your network to each other. You do that by stringing a *network cable* from the switch to each computer. The cable plugs into a special jack on the back of each computer; this jack is connected to a *network interface*, which is an electronic circuit that resides inside your computer to facilitate networking. *Voilà!* You have a working network.

If you don't want to mess with cables, you can create a wireless network instead. In a wireless network, the computers use wireless network adapters that communicate via radio signals. All modern laptop computers have built-in wireless network adapters, as do most desktop computers. (If yours doesn't, you can purchase a separate wireless network adapter that plugs into one of the computer's USB ports.) You'll need a device called a *wireless access point* (WAP) to enable the computers to properly connect. In small office or home networks, the WAP is bundled with a device called a *router*, which lets you connect your network to the internet. The combination of a WAP and a router is called a *wireless router*.

Figure 1-1 shows a typical network with five computers. This network is a home network used by a family that bears only a totally coincidental similarity to a famous TV family you may or may not have heard of. You can see that each family member has a device that connects to the network — two of them wirelessly, two of them through cables. There's also a printer that connects wirelessly.

In this example, the wireless router also has a built-in switch that provides several jacks for connecting computers via cable. Most wireless routers include this feature, typically with three to five wired network ports.

Although the network is a small one, it has much in common with larger networks that contain dozens, hundreds, or even thousands of connected computers.



**FIGURE 1-1:**  
A typical network.

Here's the rundown for each of the devices connected to this network:

- » Lisa has a laptop computer that connects wirelessly. She uses it mostly for school.
- » Bart has a fancy gaming computer that's cabled directly to the router.
- » Marge doesn't have a full-fledged computer, but she does use an iPad, which is connected wirelessly.
- » Homer has an old computer he bought at a garage sale in 1989. He doesn't know how to use it, but he doesn't know that, so no one tells him. Lisa set it up for him and repairs it when Homer breaks it (which happens every few months); she gets repair parts from eBay.
- » The printer connects wirelessly to the network and is set up so that any member of the family can print on it.
- » The wireless router connects to the internet using the family's cable TV provider. This allows everyone in the family to access the internet.

# Delving into Network Lingo

Computer networking has its own strange vocabulary. Although you don't have to know every esoteric networking term, it helps to be acquainted with a few of the basic buzzwords:



» **Local area network (LAN):** Networks are often called LANs, short for *local area network*.

LAN is the first *three-letter acronym* (TLA) of this book. You don't really need to remember it or any of the many TLAs that follow. You may guess that the acronym for *four-letter acronym* is *FLA*. Wrong! A four-letter acronym is an *ETLA*, which stands for *extended three-letter acronym*. After all, it just wouldn't be right if the acronym for *four-letter acronym* had only three letters.

» **On the network:** Every computer connected to the network is said to be "on the network." The technical term (which you can forget) for a computer that's on the network is a *node*.

» **Online, offline:** When a computer is turned on and can access the network, the computer is *online*. When a computer can't access the network, it's *offline*. A computer can be offline for several reasons. The computer can be turned off, the user may have disabled the network connection, the computer may be broken, the cable that connects it to the network can be unplugged, or a wad of gum can be jammed into the disk drive.

» **Up, down:** When a computer is turned on and working properly, it's *up*. When a computer is turned off, broken, or being serviced, it's *down*. Turning off a computer is sometimes called *taking it down*. Turning it back on is sometimes called *bringing it up*.

» **Local, remote:** A resource such as a disk drive is *local* if it resides in your computer. It's *remote* if it resides in another computer somewhere else on your network.

» **Internet:** The *internet* is a huge amalgamation of computer networks strewn about the entire planet. Networking the computers in your home or office so that they can share information with one another and connecting your computer to the worldwide internet are two separate but related tasks.

## Seeing What You Can Do with a Network

Frankly, computer networks are a bit of a pain to set up. So, why bother? Because the benefits of having a network outweigh the difficulties of setting one up.

You don't have to be a PhD to understand the benefits of networking. In fact, you learned everything you need to know in kindergarten: Networks are all about sharing. Specifically, networks are about sharing three things: files, resources, and programs.

## Accessing the internet

Probably the main reason most small business and home networks exist is to allow everyone to access the internet through a single shared internet connection. In Figure 1-1, you can see that the wireless router is connected to the internet. By sharing this connection, all the computers on the network, whether connected wirelessly or via cables, can access the internet through the wireless router.

It's important to note that nearly all wireless routers also contain a built-in *firewall*. The firewall helps protect the computers on the network from the imminent dangers of the internet. The moment you connect a home or office network to the internet, cybercriminals will begin trying to break into your network and try to trick you into divulging sensitive information, such as the password to your bank account.



WARNING

Never — and I mean *never* — allow any computer to connect directly to the internet without a firewall in place.

## Sharing files

Networks enable you to share information with other computers on the network. Depending on how you set up your network, you can share files with your network friends in several different ways. You can send a file from your computer directly to a friend's computer by attaching the file to an email message and then mailing it. Or you can let your friend access your computer over the network so that your friend can retrieve the file directly from your hard drive. Yet another method is to copy the file to a disk on another computer and then tell your friend where you put the file so that your friend can retrieve it later. One way or the other, the data travels to your friend's computer over the network cable and not on a CD or DVD or flash drive.

## Sharing resources

You can set up certain computer resources — such as hard drives or printers — so that all computers on the network can access them. For example, the printer in Figure 1-1 is a *shared resource*, which means that anyone on the network can use it. Without the network, Homer, Marge, Lisa, and Bart would have to buy their own printers.

Hard drives can be shared resources, too. In fact, you must set up a hard drive as a shared resource to share files with other users. Suppose that Lisa wants to share a file with Bart, and a shared folder has been set up on Homer's computer. All Lisa has to do is copy the file to the shared folder in Homer's computer and tell Bart where she put it. Then, when Bart gets around to it, he can copy the file from Homer's computer to his own.

## Sharing programs

Instead of keeping separate copies of programs on each person's computer, put programs on a drive that everyone shares. For example, if ten computer users all use a particular program, you can purchase and install ten copies of the program, one for each computer. Or you can purchase a ten-user license for the program and then install just one copy of the program on a shared drive. Each of the ten users can then access the program from the shared hard drive.



**WARNING**

Purchasing a single-user copy of a program and then putting it on a shared network drive — so that everyone on the network can access it — is illegal. If five people use the program, you need to either purchase five copies of the program or purchase a network license that specifically allows five or more users.



**TIP**

That being said, many software manufacturers sell their software with a concurrent usage license, which means that you can install the software on as many computers as you want, but only a certain number of people can use the software at any given time. Usually, special licensing software that runs on one of the network's server computers keeps track of how many people are currently using the software. This type of license is frequently used with more specialized (and expensive) software, such as accounting systems or computer drafting systems.

Another common method for software vendors to license their software is through a monthly or yearly subscription. You just give them your credit card number, and they give you the right to use the software. You need a working internet connection so that the software can confirm that you have a valid subscription each time you run the software.

Another benefit of networking is that networks enable computer users to communicate with one another over the network. The most obvious way networks allow computer users to communicate is by passing messages back and forth, using email or instant-messaging programs. Networks also offer other ways to communicate. For example, you can hold online meetings over the network. Network

users who have inexpensive video cameras (webcams) attached to their computers can have videoconferences. You can even play a friendly game of hearts over a network — during your lunch break, of course.

## Getting Acquainted with Servers and Clients

The network computer that contains the hard drives, printers, and other resources that are shared with other network computers is a *server*. This term comes up repeatedly, so you have to remember it. Write it on the back of your left hand.

Any computer that's not a server is a *client*. You have to remember this term, too. Write it on the back of your right hand.

Only two kinds of computers are on a network: servers and clients. Look at your left hand and then look at your right hand. Don't wash your hands until you memorize these terms.

The distinction between servers and clients in a network has parallels in sociology — in effect, a sort of class distinction between the “haves” and “have-nots” of computer resources:

- » Usually, the most powerful and expensive computers in a network are the servers. There's a good technical reason for this: All users on the network share the server's resources.
- » The cheaper and less-powerful computers in a network are the clients. *Clients* are the computers used by individual users for everyday work. Because clients' resources don't have to be shared, they don't have to be as fancy. (The exception to this rule is if the users do work that requires powerful desktop computers — for example, engineering design or video processing.)
- » Most networks have more clients than servers. For example, a network with ten clients can probably get by with one server, but larger networks will likely require more servers.
- » In most networks, a clean line of demarcation exists between servers and clients. In other words, a computer functions as either a server or a client, not both. For the sake of an efficient network, a server can't become a client, nor can a client become a server.

- » Other (usually smaller) networks can be more evenhanded by allowing any computer in the network to be a server and allowing any computer to be both a server and a client at the same time.

## Weighing Your Options: Dedicated Servers versus Peer-to-Peer Networks

In most networks, a server computer is a server computer and nothing else. It's dedicated to the sole task of providing shared resources, such as hard drives and printers, to be accessed by the network client computers. This type of server is a *dedicated server* because it can perform no other task than network services.

Some smaller networks take an alternative approach by enabling any computer on the network to function as both a client and a server. Thus, any computer can share its printers and hard drives with other computers on the network. And while a computer is working as a server, you can still use that same computer for other functions, such as word processing. This type of network is a *peer-to-peer network* because all the computers are thought of as *peers*, or equals.

Here are some points to ponder concerning the differences between dedicated-server networks and peer-to-peer networks while you're walking the dog tomorrow morning:

- » Peer-to-peer networking features are built into Windows. Thus, if your computer runs Windows, you don't have to buy any additional software to turn your computer into a server. All you have to do is enable the Windows server features.
- » The network server features that are built into Windows 11 (the most popular desktop operating system) aren't particularly efficient because this version of Windows wasn't designed primarily to be a network server.



REMEMBER

If you dedicate a computer to the task of being a full-time server, use a special server operating system rather than the standard Windows desktop operating system. A *server operating system* is specially designed to handle networking functions efficiently.

The most commonly used server operating systems are the server versions of Windows. As of this writing, the current server version of Windows is Windows Server 2025. However, many companies still use the previous version



(Windows Server 2022), and a few use even earlier versions such as Windows Server 2016 and 2019.

Another popular server operating system is Linux. Linux is popular because it's free. Linux requires more expertise to set up than Windows Server does, but it's just as capable.

- » Many networks are both peer-to-peer *and* dedicated-server networks at the same time. These networks have
  - At least one server computer that runs a server operating system such as Windows Server 2025
  - Client computers that use the server features of Windows 11 to share their resources with the network
- » Besides being dedicated, your servers should also be sincere.



TIP

## Understanding What Makes a Network Tick

To use a network, you don't really have to know much about how it works. Still, you may feel a little bit better about using the network if you realize that it doesn't work by voodoo. A network may seem like magic, but it isn't. The following list describes the inner workings of a typical network:

- » **Network interface:** Inside any computer attached to a network is a special electronic circuit called the *network interface*. The network interface has either an external jack into which you can plug a network cable — or, in the case of a wireless network interface, an antenna.
- » **Network cable:** The network cable physically connects the computers. It plugs into the network interface card (NIC) on the back of your computer.

The type of network cable most commonly used is *twisted-pair cable*, so named because it consists of several pairs of wires twisted together in a certain way. Twisted-pair cable superficially resembles telephone cable. However, appearances can be deceiving. Most phone systems are wired using a lower grade of cable that doesn't work for networks.

For the complete lowdown on networking cables, see Book 1, Chapter 2.

Network cable isn't necessary when wireless networking is used. For more information about wireless networking, see Book 1, Chapter 2.



TIP



- » **Network switch:** Networks built with twisted-pair cabling require one or more switches. A *switch* is a box with a bunch of cable connectors. Each computer on the network is connected by cable to the switch. The switch, in turn, connects all the computers to each other.

Most networks of more than a few dozen computers have more than one switch. In that case, the switches themselves are connected to each other with cable in a manner that allows all the computers to communicate with each other without regard to which switch they're directly connected to.

In the early days of twisted-pair networking, devices known as *hubs* were used rather than switches. The term *hub* is sometimes used to refer to switches, but true hubs went out of style sometime around the turn of the century.

I explain much more about switches and hubs in Book 1, Chapters 2 and 3.

- » **WAP:** In a wireless network, most cables and switches are moot. Instead, radio takes the place of cables. The device that enables a computer to connect wirelessly to a network is called a *wireless access point*. A WAP is a combination of a radio transmitter and a radio receiver and has an integrated wired network port. The WAP must be connected to the network via a cable, but it allows wireless devices such as laptops, tablets, and phones to connect wirelessly.
- » **Router:** A device found in nearly all networks is a *router*, which is used to connect two networks — typically your internal network and the internet. You can find out more about routers in Book 1, Chapters 2 and 3.
- » **Firewall:** A *firewall* is an essential component of any network that connects to the internet. The firewall provides security features that help keep cybercriminals out of your network.

In most cases, the function of a firewall is combined with the function of a router in a single device called a *firewall router*. A firewall is a security wall between two networks (usually the internet and your internal network). So, in a firewall router, the router component links the two networks, while the firewall component provides security.

In home networks or small office networks, it's also common to combine the functions of firewall, router, WAP, and switch into a single device that's usually called a *wireless router* or a *Wi-Fi router*. When you purchase such a device, check to make sure it has adequate firewall features and the correct number of switch ports for your wired devices.

# Putting the Pieces Together

In a small network such as the one that was shown in Figure 1-1, a wireless router combines the function of firewall, router, switch, and WAP. This arrangement is fine for very small networks, but when you exceed the wired switch capacity of the wireless router, you'll need additional components.

Figure 1-2 shows a network with a separate switch to connect multiple computers. Here, you can see that the wireless firewall router connects to both the internet and the switch. Several computers have wired connections to the switch, and wireless devices connect via the WAP that's built in to the Wi-Fi router. The router also provides the firewall function.

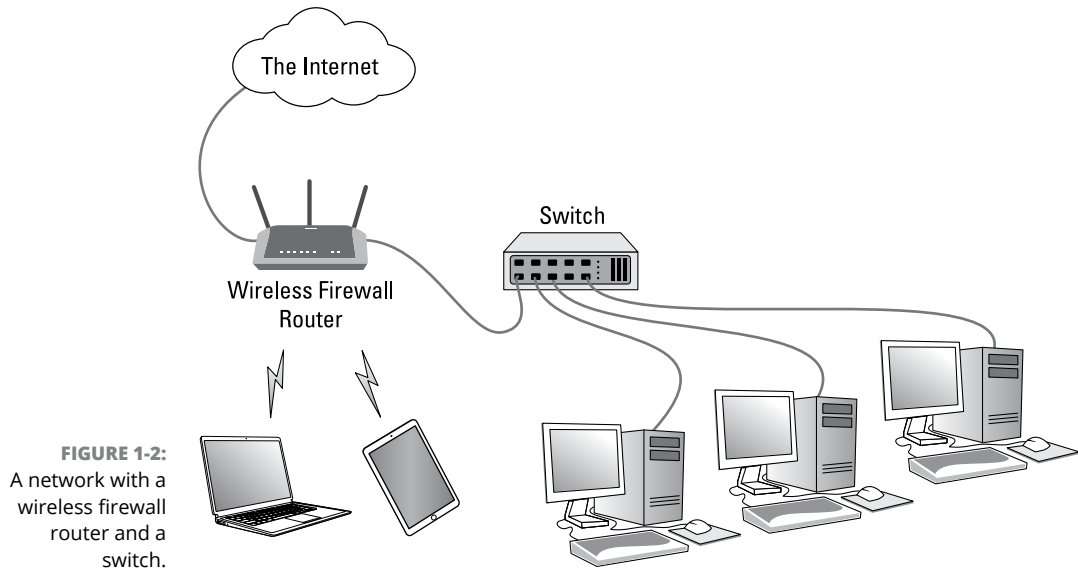
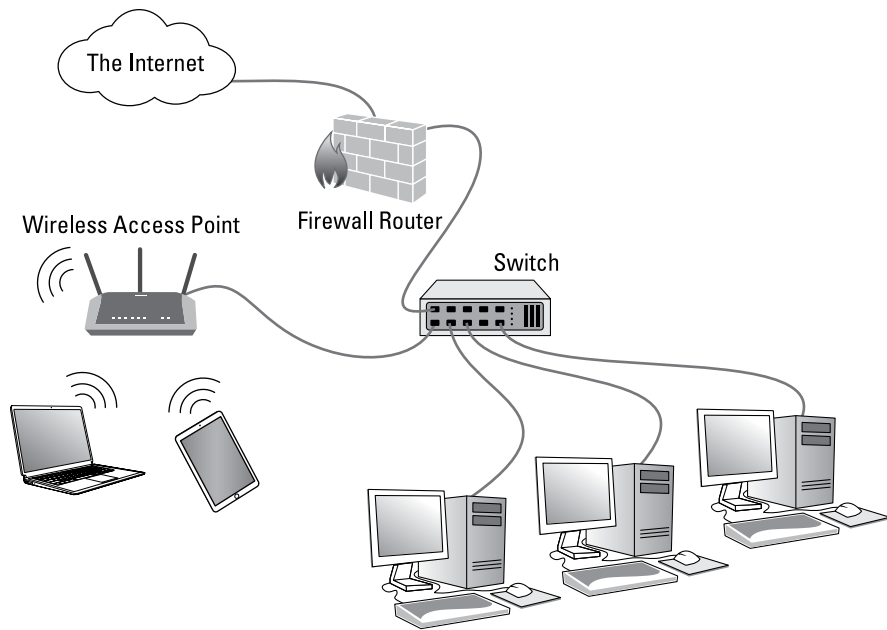


Figure 1-3 shows a more complicated setup, in which the WAP is separated from the router. Here, the router with its built-in firewall connects to the internet and to the switch. As before, several computers have wired connections to the switch. In addition, the WAP has a wired connection to the switch, allowing wireless devices to connect to the network.

In Book 1, Chapter 3, you see examples of more complicated arrangements of these basic network components.



**FIGURE 1-3:**  
A network with a  
separate firewall  
router, switch,  
and WAP.

## Considering Network Size

Networks come in all sizes and shapes. In fact, networks are commonly based on the geographical size they cover, as described in the following list:

» **LANs:** In this type of network, computers are relatively close together, such as within the same office or building.

Don't let the descriptor *local* fool you. A LAN doesn't imply that a network is small. A LAN can contain hundreds or even thousands of computers. What makes a network a LAN is that all its connected computers are located within close proximity. Usually a LAN is contained within a single building, but a LAN can extend to several buildings on a campus, provided that the buildings are close to each other (typically within 300 feet, although greater distances are possible with special equipment).

» **Wide area networks (WANs):** These networks span a large geographic territory, such as an entire city, a region, or even a country. WANs are typically used to connect two or more LANs that are relatively far apart. For example, a WAN may connect an office in San Francisco with an office in New York.



REMEMBER

The geographic distance, not the number of computers involved, makes a network a WAN. If an office in San Francisco and an office in New York each has only one computer, the WAN will have a grand sum of two computers — but will span more than 3,000 miles.

- » **Metropolitan area networks (MANs):** This kind of network is smaller than a typical WAN but larger than a LAN. Typically, a MAN connects two or more LANs within the same city that are far enough apart that the networks can't be connected via a simple cable or wireless connection.

## Recognizing That Your Personal Computer Isn't Personal When It's on a Network

If I had to choose one point that I want you to remember from this chapter more than anything else, it's this: After you hook up your personal computer (PC) to a network, it's not a "personal" computer anymore. You're now part of a network of computers, and in a way, you've given up one of the key concepts that made PCs so successful in the first place: independence.

I got my start in computers back in the days when mainframe computers ruled the roost. *Mainframe computers* are big, complex machines that used to fill entire rooms and had to be cooled with chilled water. My first computer was a water-cooled Acme Hex Core Model 2000. (I'm not making up the part about the water. A plumber was often required to install a mainframe computer. In fact, the really big ones were cooled by liquid nitrogen. I *am* making up the part about the Acme Hex Core 2000.)

Mainframe computers required staffs of programmers and operators in white lab coats just to keep them going. The mainframes had to be carefully managed. A whole bureaucracy grew up around managing them.

Mainframe computers used to be the dominant computers in the workplace. PCs changed all that: They took the computing power out of the big computer room and put it on the user's desktop, where it belongs. PCs severed the tie to the centralized control of the mainframe computer. With a PC, a user could look at the computer and say, "This is mine — all mine!" Mainframes still exist, but they're not nearly as popular as they once were.

But networks have changed everything all over again. In a way, it's a change back to the mainframe-computer way of thinking: central location, distributed resources. True, the network isn't housed in the basement and doesn't have to be installed by a plumber. But you can no longer think of "your" PC as your own. You're part of a network — and like the mainframe, the network has to be carefully managed.

Here are several ways in which a network robs you of your independence:

- » **You can't just indiscriminately delete files from the network.** They may not be yours.
- » **You're forced to be concerned about network security.** For example, a server computer has to know who you are before it allows you to access its files. So, you have to know your user ID and password to access the network. This precaution prevents some 15-year-old kid from hacking their way into your office network by using its internet connection and stealing all your computer games.
- » **You may have to wait for shared resources.** You may need to print a quick page on your way into a meeting that you're already late for, only to discover that someone else sent a 1,000-page document to the printer. You'll have to wait or find a different printer.
- » **You may have to wait for access to documents.** You may try to retrieve a Microsoft Excel spreadsheet file from a network drive, only to discover that someone else is using it. You'll just have to wait. (Newer technologies have made it possible for multiple people to edit files at the same time, which is kind of mind-blowing.)
- » **You don't have unlimited storage space.** If you copy a 100GB video file to a server's drive, you may get calls later from angry coworkers complaining that no room is left on the server's drive for their important files.
- » **Your files can become infected by viruses given to you by someone over the network.** You may then accidentally infect other network users.
- » **You have to be careful about saving sensitive files on the server.** If you write an angry note about your boss and save it on the server's hard drive, your boss may find the memo and read it.
- » **The server computers may be down for maintenance.** This happens all the time. Servers need to be kept up to date with system updates, or new software may need to be installed. At times, the servers will be taken offline for such purposes. When the servers are offline, you'll have to wait. (Most IT administrators schedule server downtime at weird hours, like midnight, so these outages shouldn't affect you unless you keep odd hours.)

# Assigning a Network Administrator

Because so much can go wrong — even with a simple network — designating one person as network administrator is important. This way, someone is responsible for making sure that the network doesn't fall apart or get out of control.

The network administrator doesn't have to be a technical genius. In fact, some of the best network administrators are complete idiots when it comes to technical stuff. What's important is that the administrator is organized. That person's job is to make sure that plenty of space is available on the file server, that the file server is backed up regularly, and that new employees can access the network, among other tasks.

The network administrator's job also includes solving basic problems that the users themselves can't solve — and knowing when to call in an expert (when something really bad happens). It's a tough job, but somebody's got to do it. Here are a few tips that may help:

- » In small companies, picking the network administrator by drawing straws is common. The person who draws the shortest straw loses and becomes administrator.
- » Of course, the network administrator can't be a *complete* technical idiot. I was lying about that. (For those of you in Congress, the word is *testifying*.) I exaggerated to make the point that organizational skills are more important than technical skills. The network administrator needs to know how to do various maintenance tasks. Although this knowledge requires at least a little technical know-how, the organizational skills are more important.

Because network administration is such an important job, all the chapters in Book 9 are devoted to it.

